EMBEDDING GENERATION

In this step, we'll convert a list of IT support questions into embeddings and store them in memory.

Listing 5.9 Converting text to embeddings

```
const embeddingDB = {};
const questions = [
  "How do I reset my password?",
  "What should I do if my computer won't start?",
];

const answers = [
  "To reset your password, go to the login page and click
'Forgot Password'. Follow the instructions to reset
your password.",
  "If your computer won't start, check the power cable, try
restarting it, and if the issue persists, contact support.",
];

for (let i = 0; i < questions.length; i++) {
  const { embedding } = await embed({
    model: google.textEmbeddingModel("text-embedding-004"),
    value: questions[i],
  });
  embeddingDB[questions[i]] = embedding;
}
```

This line defines an object (embeddingDB) to serve as an in-memory database, which stores the question embeddings.

Defines an array of sample questions for the IT support knowledge base

Defines an array of corresponding answers to the questions defined in the "questions" array

This line saves each generated embedding in the embeddingDB object, using the question as the key and the embedding as the value.

Here, we loop through the questions array, generate embeddings using OpenAI's embedding model, and store each embedding in an in-memory object (`embeddingDB`) with the question as the key. We use an in-memory object instead of a vector database because it's not important at this point, as it will complicate the example.

Next, we allow the application to accept user queries; we use the `embed` function to generate an embedding for the query.

USER QUERY

When a user asks a question, the system generates an embedding for that query. We simply use the embed function to get the embeddings of the query.

Listing 5.10 Embedding the user query

```
const userQuery = "I forgot my password";
const { embedding: queryEmbedding } = await embed({
  model: google.textEmbeddingModel("text-embedding-004"),
  value: userQuery,
});
```